

Maypole Braids: An Analysis Using the Annular Braid Group

Moyi Tian & Professor David Richeson
Department of Mathematics and Computer Science
Dickinson College

Dickinson

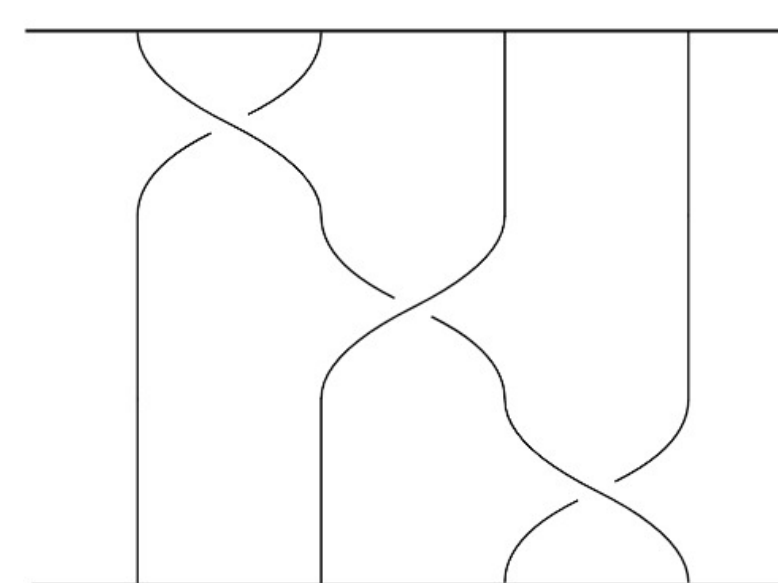
Background

In 1926, Emil Artin revealed the natural group structure in ordinary braids by introducing braid groups.

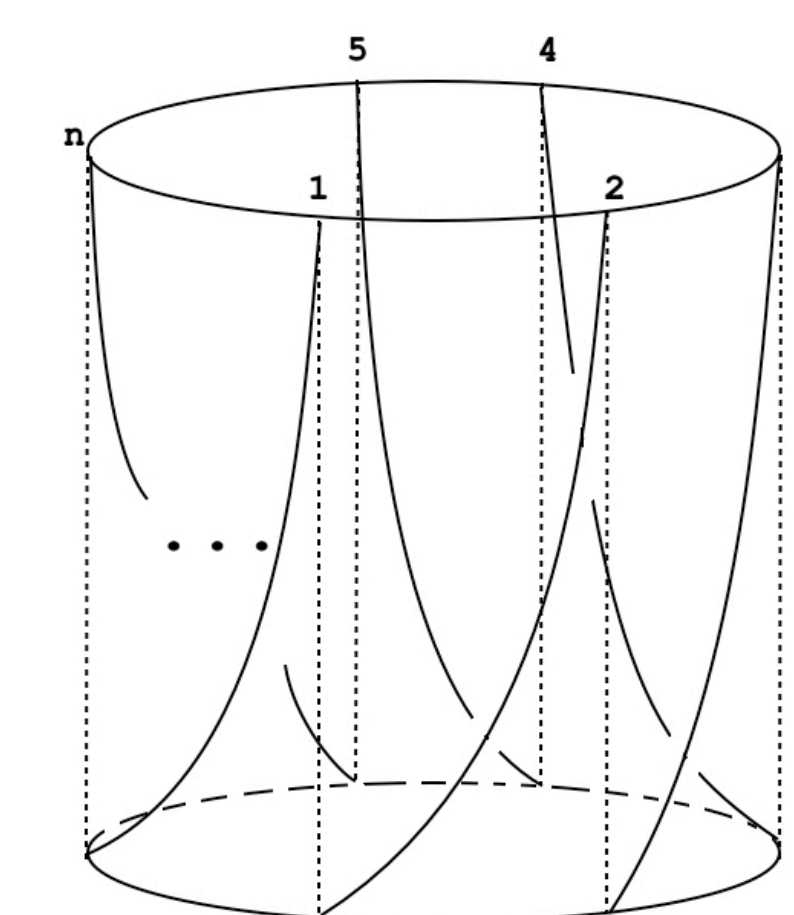
The **Artin braid group** B_n is a group generated by $n-1$ generators $s_1, s_2, \dots, s_{n-1}$ and the **braid relations**

- $s_i s_j = s_j s_i$ for all $i, j = 1, 2, \dots, n-1$ with $|i-j| \geq 2$, and
- $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ for $i = 1, 2, \dots, n-2$.

All algebraic braids have corresponding geometric presentations. We can describe a braid by listing the crossing-overs of strands, which are denoted by the generators. s_i is the i^{th} string crossing over the $(i+1)^{\text{st}}$ string and s_{i-1} is the $(i+1)^{\text{st}}$ string crossing over the i^{th} string.



Braid $s_1 s_2 s_3$



The **Annular braid** is similar to the ordinary braid except that the top and bottom are circular and there is a pole down the center. We allow the first and the last strand to twist with each other as neighbors. Also, all strands can twist simultaneously without crossing over one another, which we call the rotation generator τ .



Maypole dances are performed in some European festivals. A maypole is a wooden pole with ribbons attached to its top. In a maypole dance, each person holds a ribbon and follows some rules to move in a circle. The ribbons gradually braid around the pole and finally form a special decoration with an interesting pattern. The created patterns are analogous to annular braids.

Presentations

There are multiple ways to present the annular braid group.

1

Annular braids with n strands. σ_i corresponds to the i^{th} strand crossing over the $(i+1)^{\text{st}}$ strand. τ is the rotation. Generators are indexed modulo n .

$$P = \langle \tau, \sigma_0, \sigma_1, \sigma_2, \dots, \sigma_{n-1} \mid \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \text{ for } i = 0, 1, \dots, n-1, \tau \sigma_i \tau^{-1} = \sigma_{i+1} \text{ for } i = 0, 1, \dots, n-1 \rangle.$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i-j| \neq 1, n-1, \tau \sigma_i \tau^{-1} = \sigma_{i+1} \text{ for } i = 0, 1, \dots, n-1.$$

2

Let the first strand be the pole and give it the property that it always ends in the first spot at the bottom.

$$\mathcal{D} = \langle \gamma_2, \gamma_3, \dots, \gamma_n, a_2, a_3, \dots, a_{n+1} \mid \gamma_i \gamma_{i+1} \gamma_i = \gamma_{i+1} \gamma_i \gamma_{i+1} \text{ for } i = 2, 3, \dots, n-1, \gamma_i \gamma_j = \gamma_j \gamma_i \text{ for } |i-j| \neq 1, n-1, \gamma_i a_k \gamma_i^{-1} = a_k \text{ for } k \neq i, i+1, \gamma_i a_i \gamma_i^{-1} = a_{i+1}, \gamma_i a_{i+1} \gamma_i^{-1} = a_{i+1}^{-1} a_i a_{i+1} \text{ for } i = 2, 3, \dots, n \rangle.$$

$$\gamma_i \gamma_j = \gamma_j \gamma_i \text{ for } |i-j| \neq 1, n-1, \gamma_i a_k \gamma_i^{-1} = a_k \text{ for } k \neq i, i+1, \gamma_i a_i \gamma_i^{-1} = a_{i+1}, \gamma_i a_{i+1} \gamma_i^{-1} = a_{i+1}^{-1} a_i a_{i+1} \text{ for } i = 2, 3, \dots, n.$$

$$\gamma_i a_i \gamma_i^{-1} = a_{i+1}, \gamma_i a_{i+1} \gamma_i^{-1} = a_{i+1}^{-1} a_i a_{i+1} \text{ for } i = 2, 3, \dots, n.$$

Let $\sigma_2, \sigma_3, \dots, \sigma_n$ be $\gamma_2, \gamma_3, \dots, \gamma_n$ and $a_i = \sigma_i^{-1} \sigma_{i-1}^{-1} \dots \sigma_{i-2}^{-1} \sigma_{i-1} \sigma_{i-2} \dots \sigma_2 \sigma_1$. Generators are indexed modulo $n+1$.

The effect of σ_0 can be achieved by

$$a_2^{-1} \gamma_2^{-1} \gamma_3^{-1} \dots \gamma_{n-1}^{-1} \gamma_n^{-1} \gamma_n \gamma_{n-1} \gamma_{n-2} \dots \gamma_3 \gamma_2 a_2$$

The effect of τ can be achieved by

$$a_2^{-1} \gamma_2^{-1} \gamma_3^{-1} \dots \gamma_{n-1}^{-1} \gamma_n^{-1}$$

$\Leftarrow \sigma_0$

3

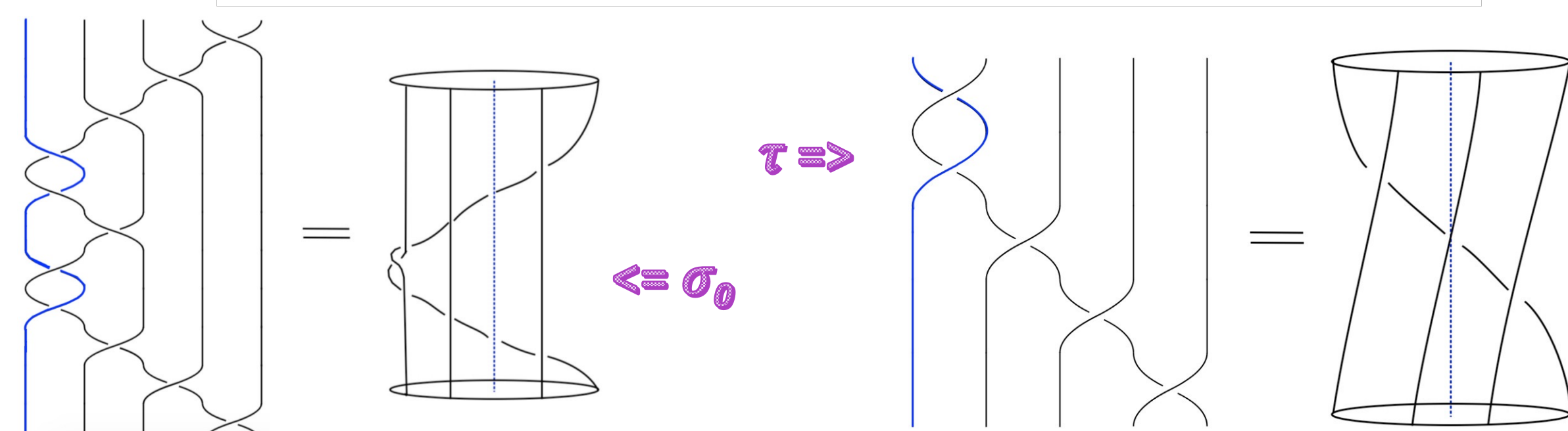
First strand is the pole. Let $\Sigma_i = \sigma_i$ for $i=2, 3, \dots, n$. Generators are indexed modulo $n+1$.

$$\Sigma_0 = \sigma_n \sigma_{n-1} \dots \sigma_2 \sigma_1^{-1} \sigma_2 \sigma_1^{-1} \dots \sigma_{n-1}^{-1} \sigma_n^{-1} \text{ and } T = \sigma_1^{-1} \sigma_2^{-1} \dots \sigma_{n-1}^{-1} \sigma_n^{-1}$$

$$\mathcal{C} = \langle \Sigma_0 \Sigma_2, \Sigma_3, \dots, \Sigma_n, T \mid \Sigma_i \Sigma_j = \Sigma_j \Sigma_i \text{ for all } i, j = 0, 2, 3, \dots, n \text{ with } |i-j| \geq 2, \text{ except that } \Sigma_2 \Sigma_0 \neq \Sigma_0 \Sigma_2, \Sigma_i \Sigma_{i+1} \Sigma_i = \Sigma_{i+1} \Sigma_i \Sigma_{i+1} \text{ for } i = 2, 3, \dots, n, \text{ and } \Sigma_0 \Sigma_2 \Sigma_0 = \Sigma_2 \Sigma_0 \Sigma_2, T \Sigma_i T^{-1} = \Sigma_{i+1} \text{ for } i = 2, 3, \dots, n \text{ and } T \Sigma_0 T^{-1} = \Sigma_2 \rangle.$$

$$\text{except that } \Sigma_2 \Sigma_0 \neq \Sigma_0 \Sigma_2, \Sigma_i \Sigma_{i+1} \Sigma_i = \Sigma_{i+1} \Sigma_i \Sigma_{i+1} \text{ for } i = 2, 3, \dots, n$$

$$\text{and } \Sigma_0 \Sigma_2 \Sigma_0 = \Sigma_2 \Sigma_0 \Sigma_2, T \Sigma_i T^{-1} = \Sigma_{i+1} \text{ for } i = 2, 3, \dots, n \text{ and } T \Sigma_0 T^{-1} = \Sigma_2.$$

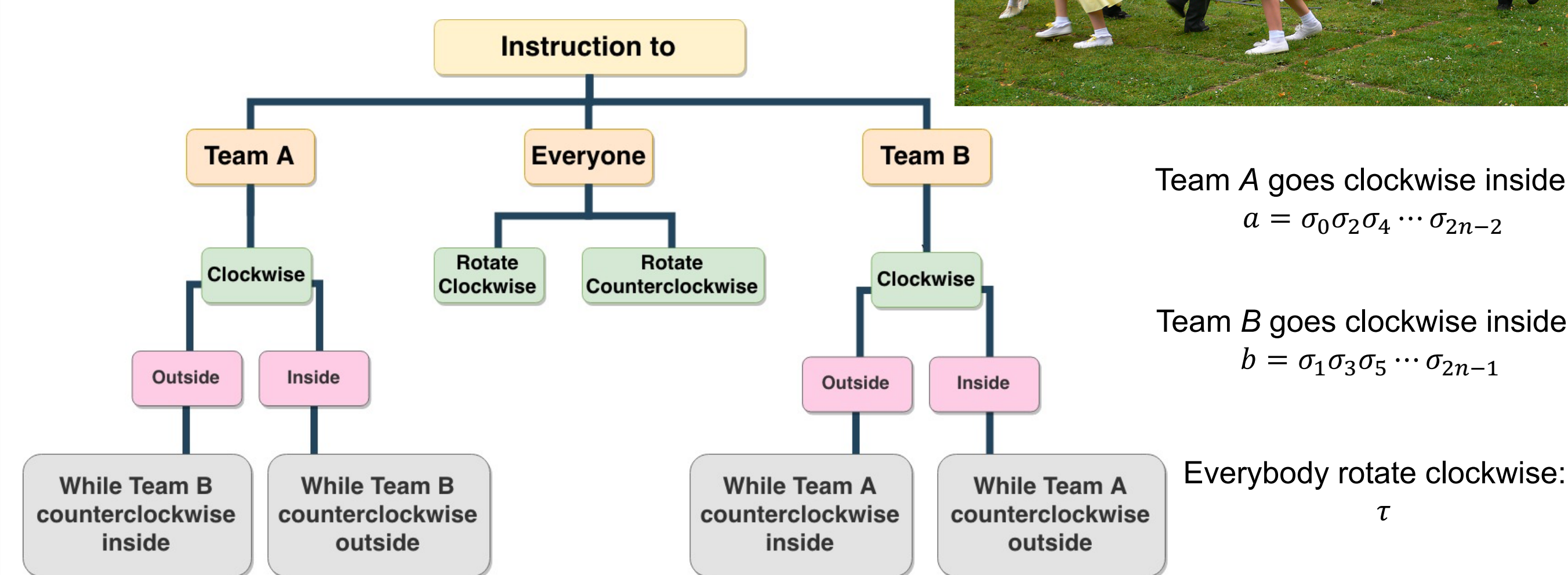


$\Leftarrow \sigma_0$

Application to Maypole Dances

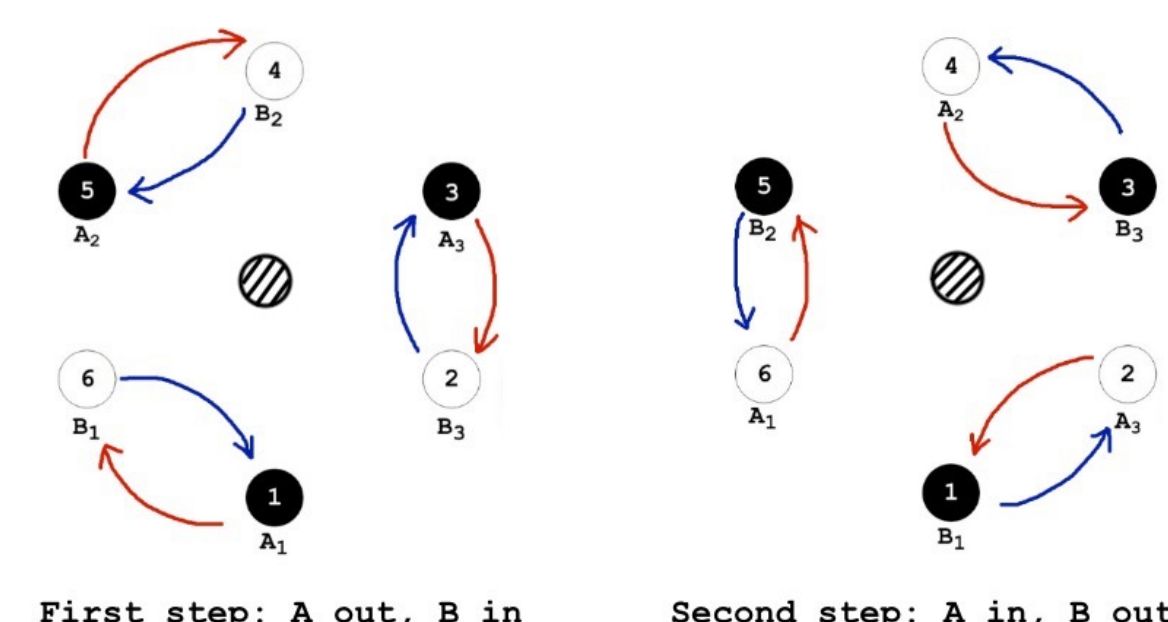
Maypole dances can be understood at three different levels - the sequence of instructions to each dancer, the sequence of overhead moves and the braid diagram.

In a dance of two teams with $2n$ people, there are three generators of the fundamental dance moves.



The Grand Chain

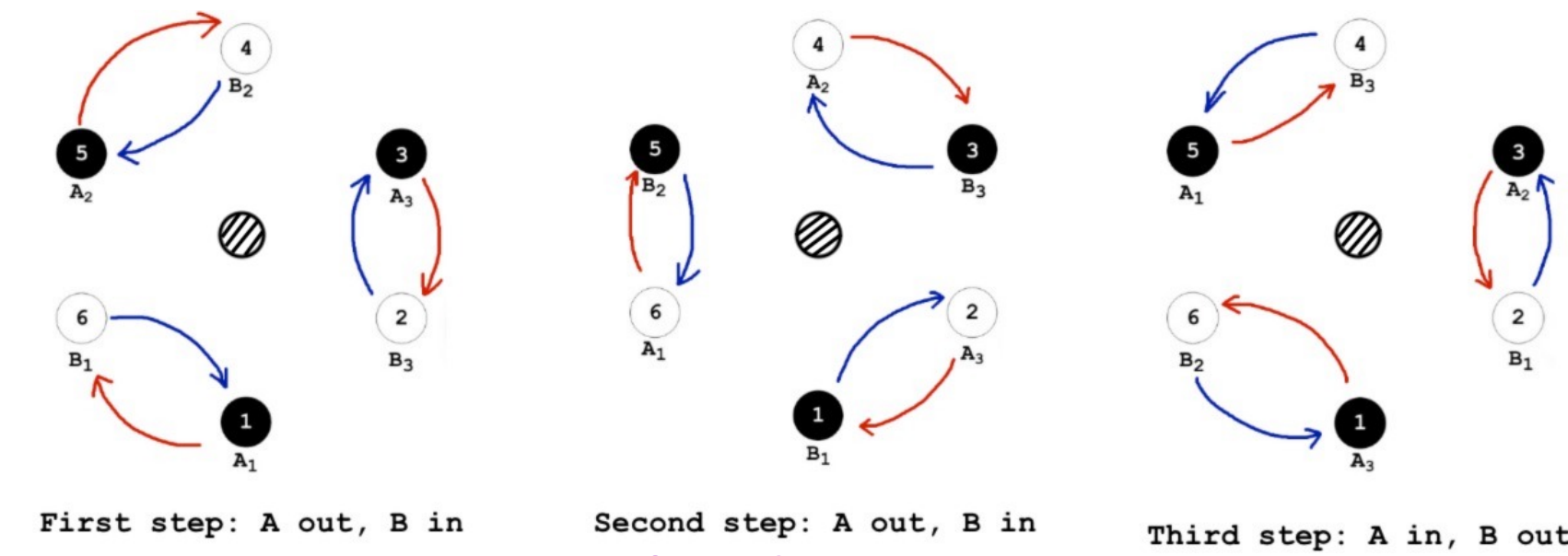
The instruction of this dance is $a^{-1}b$



Overhead moves

A derivation:

The instruction is $a^{-1}b^{-1}ab^{-1}a^{-1}b$



Overhead moves

Braid diagram

Braid diagram

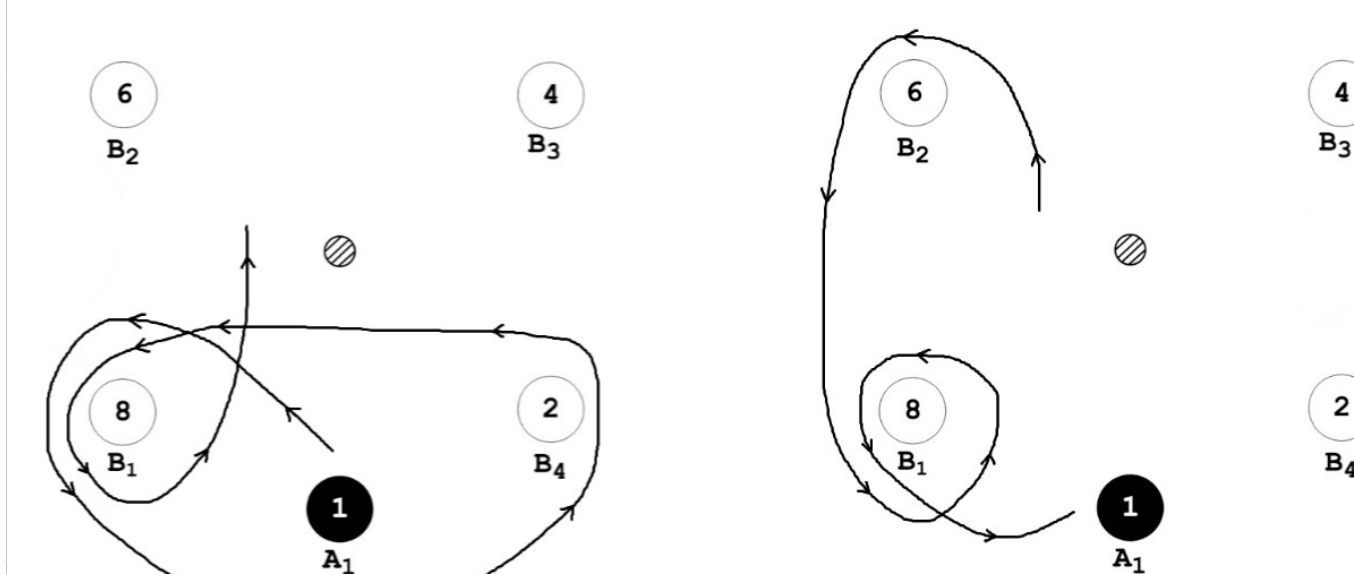
The Gypsies' Tent

The word of this braid:

$$a\tau(b\tau^{-1})^2(a\tau)^2b\tau^{-1}(a\tau)^2(b\tau^{-1})^2a\tau b\tau^{-1}$$

The generators are:

$$a\tau \text{ and } b\tau^{-1}$$



Step 1

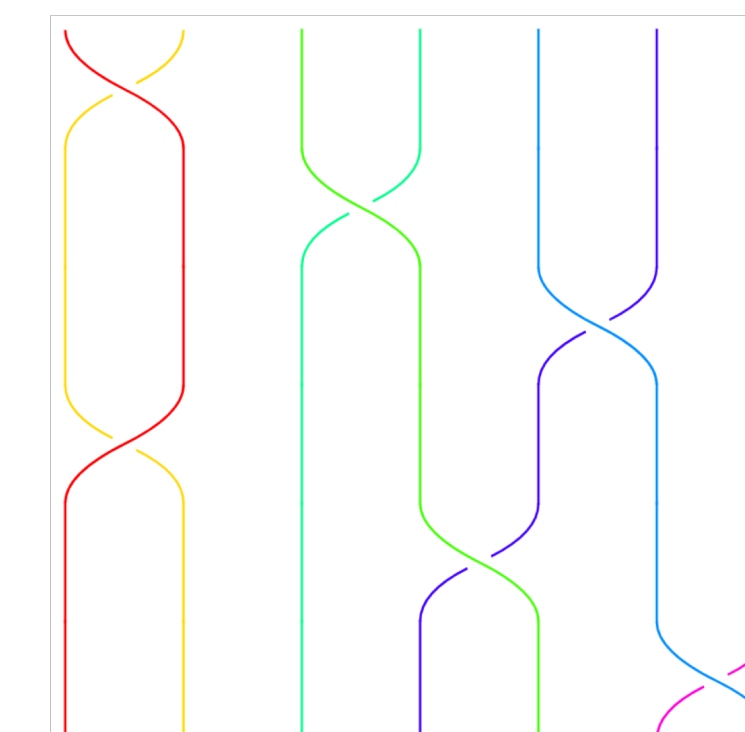
Step 2

Partial illustration of overhead moves

$\Leftarrow$ Braid diagram

Invariants

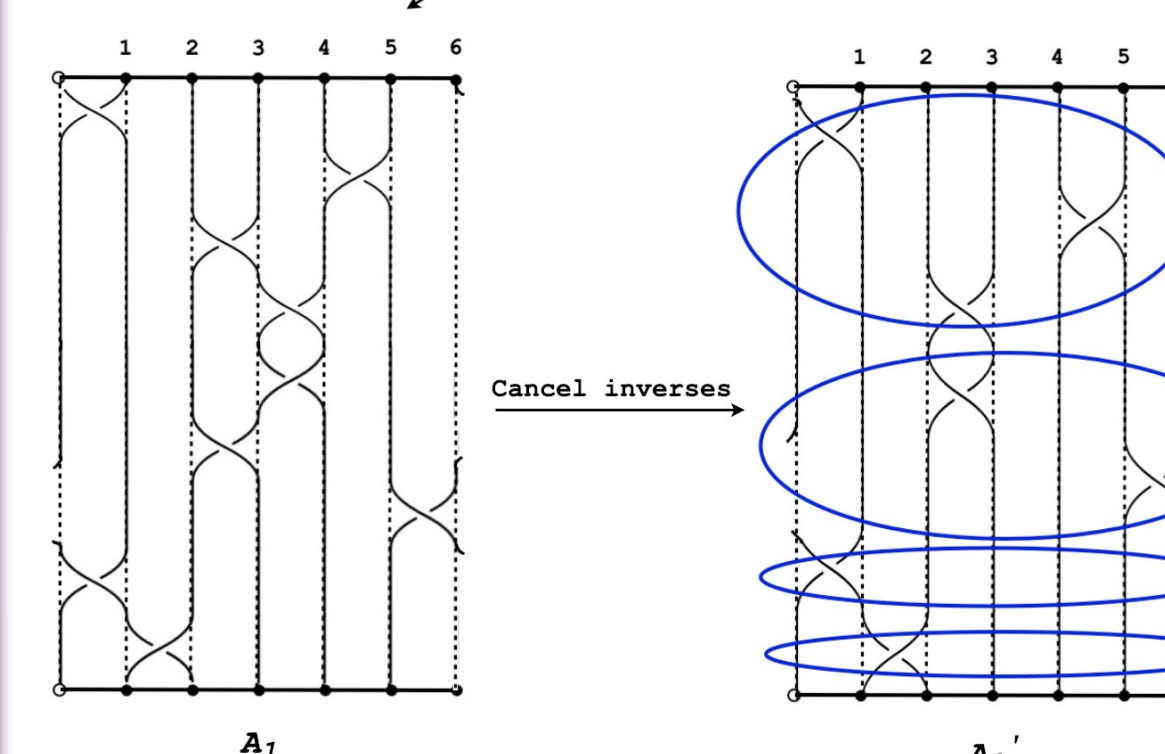
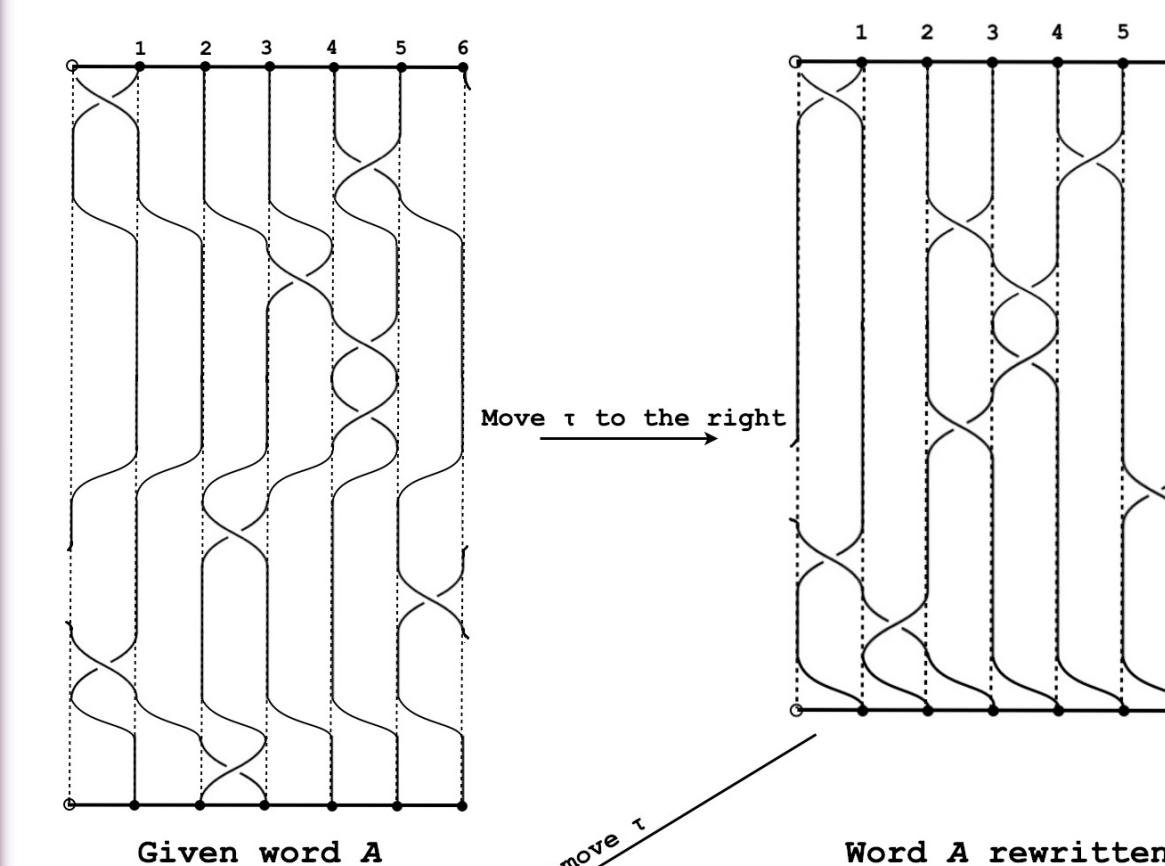
Minimum Number of Crossings



What is the least number of crossings required to get us the same braid? In annular braid group, all rotational terms can be pushed to the end by applying braid relations and the question turns out to be the shortest word problem in ordinary braid group. Finding the shortest word of a braid in B_n is an NP-complete problem.

Greedy Algorithm: Given a finite braid in B_n , if we do not allow the word length to increase, we can do an exhaustive search for all possible transformations. For each of them, we cancel the neighboring inverses and find out the number of crossings. The local minimum is the lowest number among these numbers of crossings.

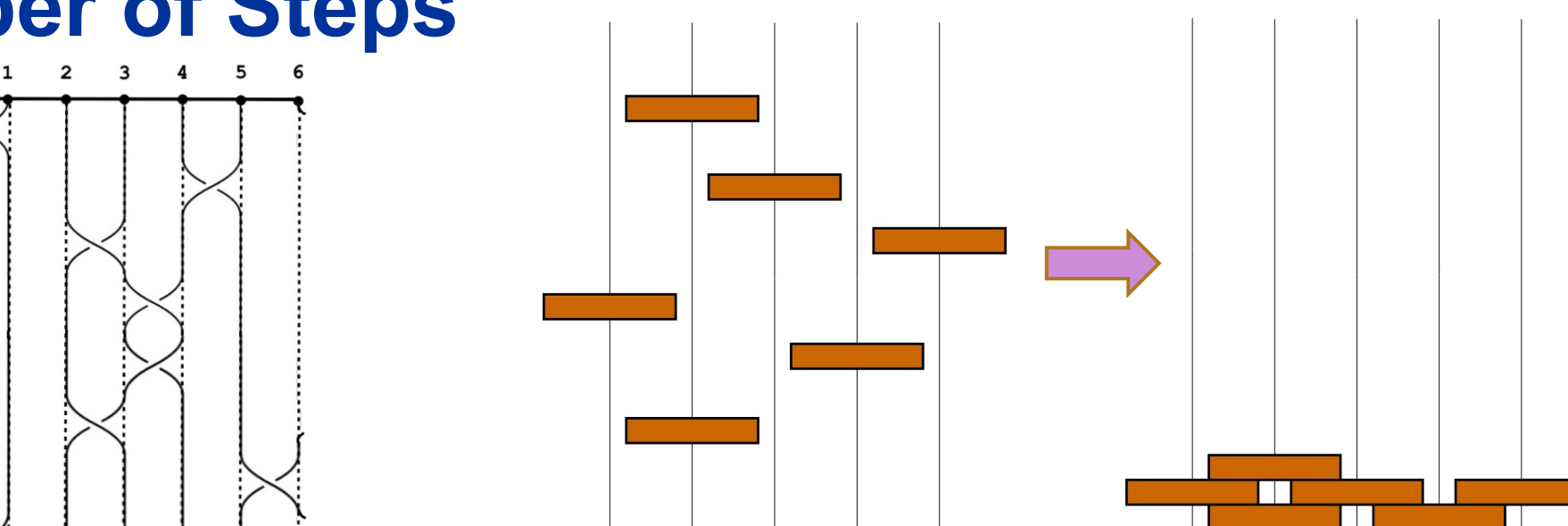
Minimum Number of Steps



Transformations of a braid word yield different numbers of steps.

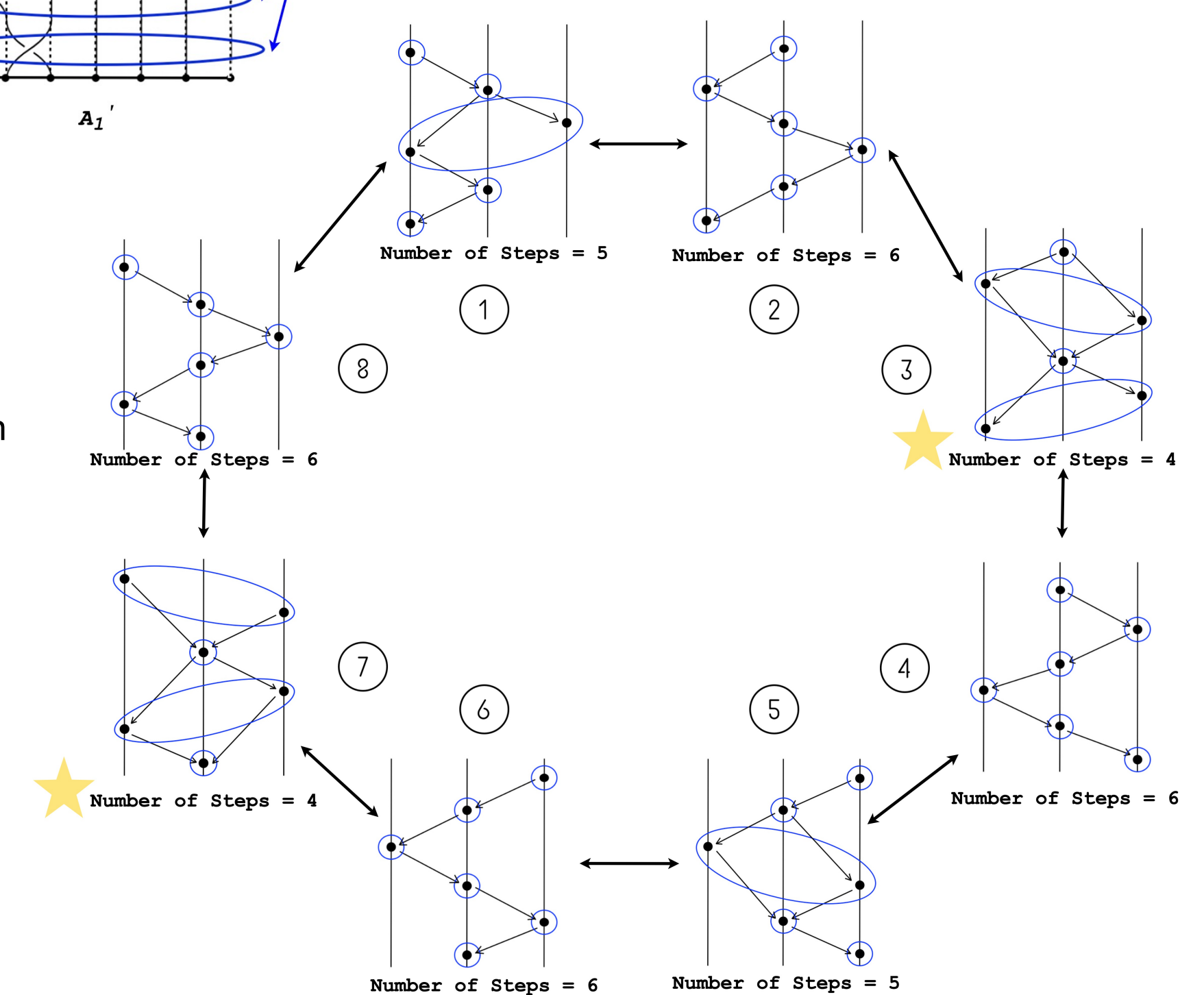
Algorithm (for local minimum):

- 1) Use braid relations to move all the rotation generators to the right and take the part purely with σ -generators to work with.
- 2) Apply the greedy algorithm to find out the braid word with minimal length.
- 3) Do an exhaustive search on the shortest word. Apply the method of directed graph to figure out the number of steps.
- 4) The minimum among these numbers is the local minimum number of steps.



Like the toy blocks on rigid poles falling by gravity and piling up with several layers due to their spacial relations, each step of a dance can have several people moving simultaneously. We are interested in figuring out the minimum number of steps required for getting us the same braid.

We can again turn this into a problem for ordinary braids. Each braid word has an associated number of groupings. A procedure is to draw directed graphs. To find the global minimum, we sometimes have to go worse in order to achieve the optimal.



Conclusion and Future Work

Three presentations of the annular braid group are provided in this research. Each maypole dance can be viewed by instructions, the overhead moves and the braid diagram. The annular braid group can serve as a tool to algebraically analyze traditional maypole dances. We can use the generators to give instructions more efficiently and also construct new dances from the braid group. The maypole dances revealed invariants of the annular braid group - the minimum number of crossings and the minimum number of steps. We proposed algorithms to find the local optimal solution. Future research can focus on giving an algorithm that yields a better approximation or finding the global optimal solution. The related shortest word problem is known as NP-complete. The proof of properties of the braid group can be helpful to the applications in cryptography.

REFERENCES

- [1] Bangert, P. D. (2009). Braids and Knots. In R. L. Ricca (Ed.), *Lectures on Topological Fluid Mechanics*, Lecture Notes in Mathematics, (1-73). Springer, Berlin, Heidelberg.
- [2] Kassel, C., Dodane, O., & Turaev, V. (2008). *Braid Groups*. New York, NY: Springer.
- [3] Kent, R. P., & Peifer, D. (2002). "A Geometric and Algebraic Description of Annular Braid Groups", *International Journal of Algebra and Computation*, 12(1/2), 85-97.